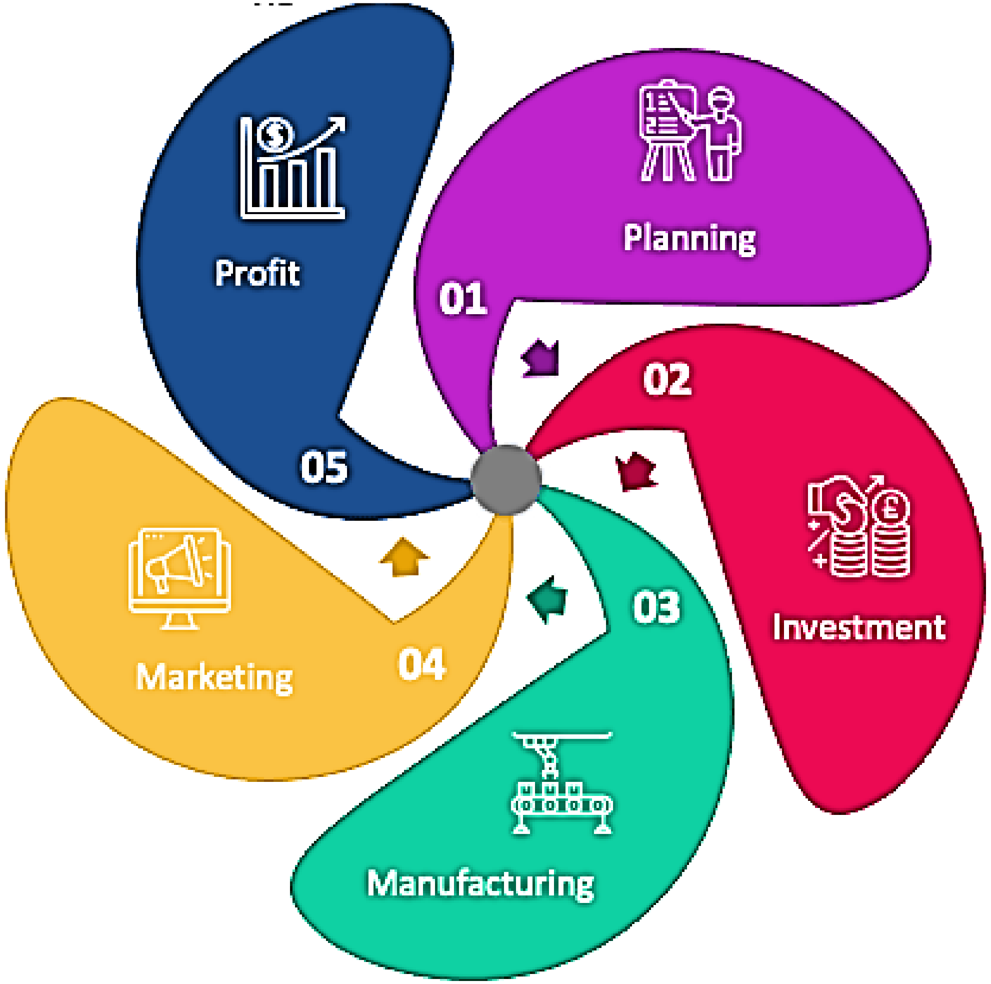

بسم الله الرحمن الرحيم
In the name of Allah, Most Gracious,
Most Merciful.

Course HS - 402, Engineering Economics, CR 1, (2+0)/week

Engineering Economics (HS - 407)



Course Instructor

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Course Outline & Reference Books

Join Google Classroom:

https://classroom.google.com/c/NzYwNDU5NjkwNDg2?cjc=rk4peiq

rk4peiq

Course Outline & Reference Books

Reference Books

1. Engineering Economy by William G. Sullivan, Elin W. Wicks, C. Patrick Koelling

2. Economics by Campbell R Cornell, Stanley L. Bruce

3. Project Management, the managerial process by Erik W. Larson, Clifford F. Gray

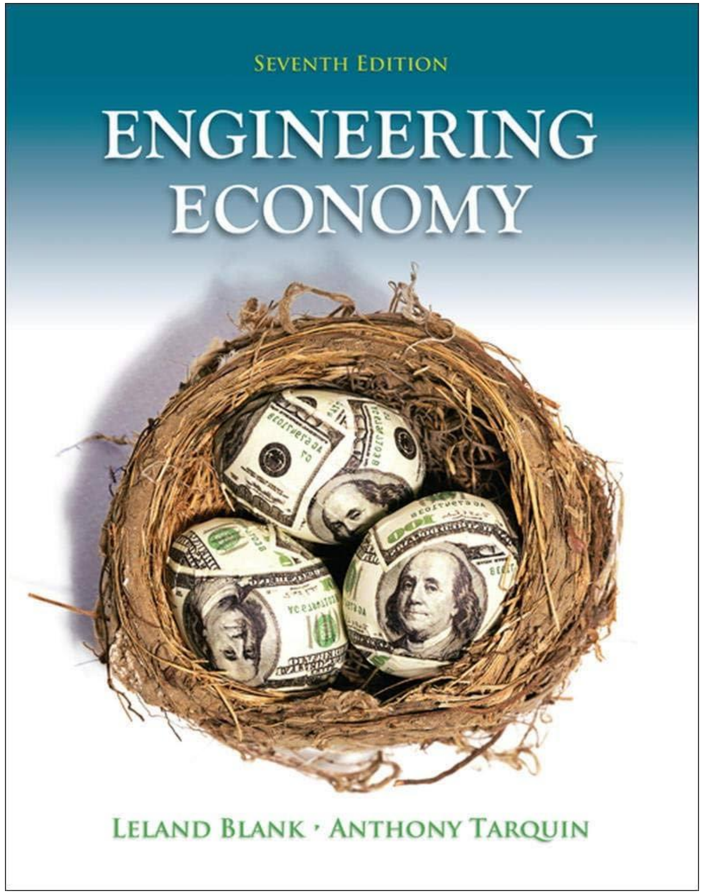
4. Karl E. Case's Principles of Economics (10th Edition)

Office Visiting Hours

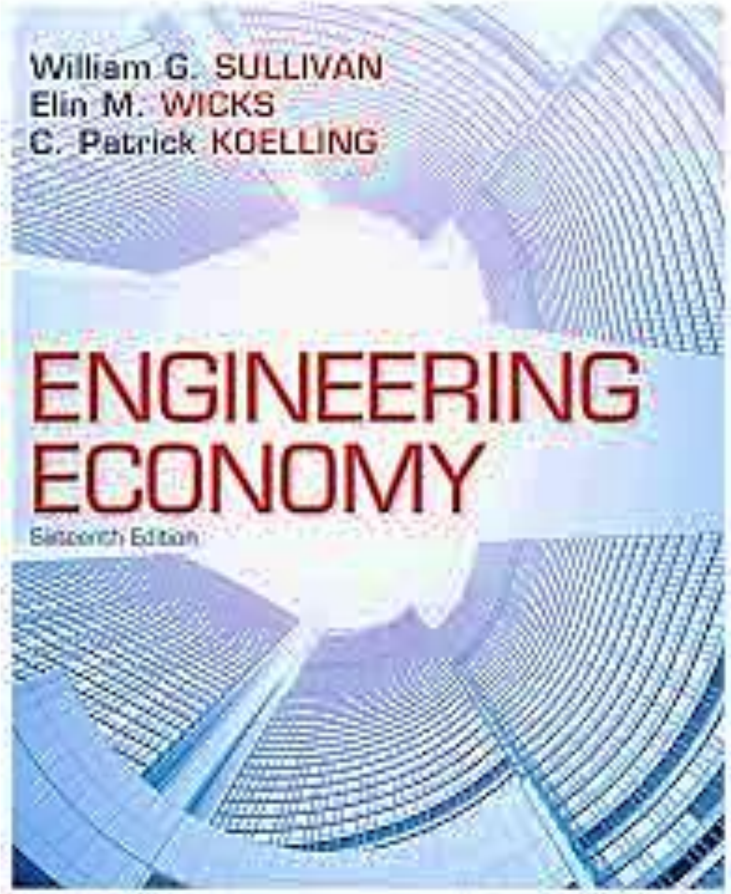
- Office 08 MED,
- Phone: Ext 327/329
- Office Hours: 0830 - 1530 hrs.
- Lunch Break 1230 - 1330 hrs.
- Student Hours: 0930 - 1630 hrs. Tuesday

Course Outline & Reference Books

Reference Book 1



Reference Book 2



Reference Book 3

Course HS - 402, Engineering Economics, CR 1, (2+0)/week

Course Outline & Grading Policy

Assignments=10% Quizzes=05% for 1/CH → 2 Quizzes, Project=10% Mid Exam=30% each, Finals=45%

Quiz 1 . . . . .Week 03
Assignment 1 . . . . .Week 04
Quiz 2. . . . . .Week 05
Assignment 2 . . . . .Week 06

Mid Exam . . . . Week 08
Quiz 3 . . . . .Week 10
Assignment 3 . . . . .Week 11
Quiz 4 . . . . .Week 12
Assignment 4 . . . . .Week 13
Project . . . . .Week 14
Presentation. . . . . .Week 15

Finals . . . . Week 17

Course Outline & Class Policy

- You are highly encouraged to ask questions and discuss the course material with me and your classmates.
- No cheating, fabrication, falsification, forgery and computer misuse.
- Please switch off/put on silent your cell phones.
- Adhere to the given date for assignment submission.
- Academic Honesty: Plagiarism will not be tolerated at any level.
- Extra Help: Do not hesitate to come to my office during office hours or by appointment to discuss a problem or any aspect of the course.
- University Attendance Policy: Students are expected to attend classes regularly. Minimum attendance must be 75%.

Course Outline & Important Note

Make-Up Of Any Exams / Quizzes / Sessional Will Not Be Entertained – Make Sure To Attend It As Scheduled

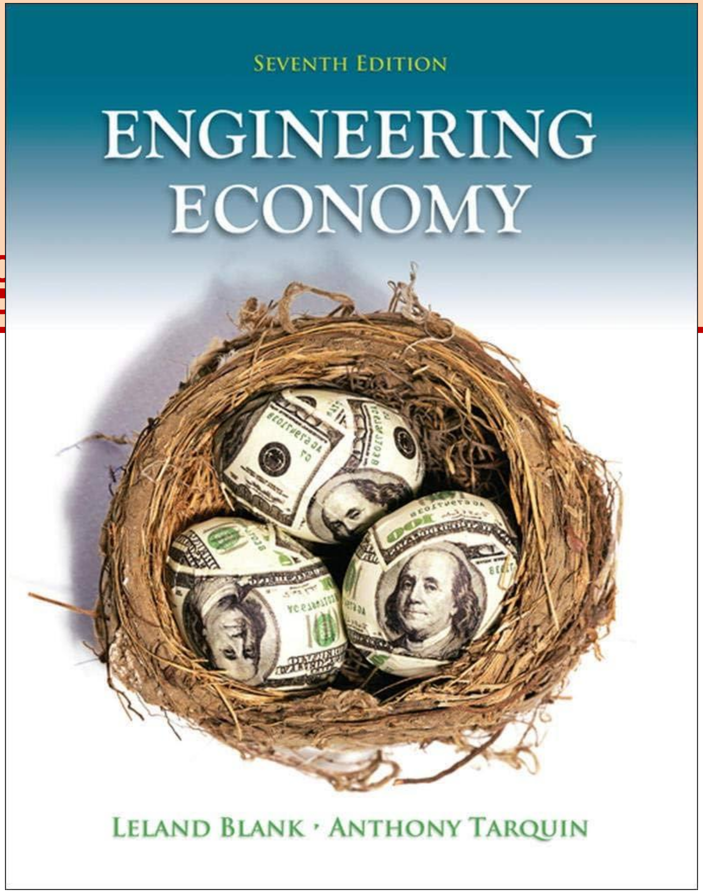
Make Sure To Comply With The Attendance Policy (75% minimum). No Relaxation Will Be Granted

Make Habit of Self-Study – Make Habit of Reading Books
Make Use of the Power of Internet for Research

Introduction to Engineering Economics

Chapter # (1)

The Scope and Nature of Engineering



The Foundation of Engineering Economics by Leland Blank

Introduction to Engineering Economics

- Scarcity exists because individuals want more than can be produced
- Scarcity means the goods available are too few to satisfy individuals' desires.
- The degree of scarcity is constantly changing
- The quantity of goods, services and Usable Resources depends on technology and human action

The Economic Problem: Scarcity & Choice

Economics is the study of choices when there is scarcity.

Here are some examples of scarcity and the trade-offs associated with making choices:

- You have a limited amount of time. If you take a part-time job, each hour on the job means one less hour for study or play.
- A city has a limited amount of land. If the city uses an acre of land for a park, it has one less acre for housing, retailers, or industry.
- You have limited income this year. If you spend $17 on a music CD, that's $17 less you have to spend on other products or to save.

Introduction to Engineering Economics

Economics Objectives

The objective is the goal or desired outcome of a choice

- **Individuals may try to maximize utility given the constraints of income, time, prices, etc.**
- **Firms may have objectives such as the maximization of profits, sales, market share, etc. or the minimization of costs per unit**
- **Social objective, maximize the well being of the members of society**

Microeconomics studies such things as:

The pricing policy of firms

Household's decisions on what to buy

How markets allocate resources among alternative ends

Macroeconomics studies such things as:

Inflation

Unemployment

Economic growth

Introduction to Engineering Economics
(Scope of Economics)

Examples of microeconomic and macroeconomic concerns				
	Production	**Prices**	**Income**	**Employment**
Microeconomics	Production/Output in Individual Industries and Businesses ❖ How much steel ❖ How many offices ❖ How many cars	Price of Individual Goods and Services ❖ Price of medical care ❖ Price of gasoline ❖ Food prices ❖ Apartment rents	Distribution of Income and Wealth ❖ Wages in the auto Industry ❖ Minimum wages ❖ Executive salaries ❖ Poverty	❖ Employment by Individual Businesses & Industries ❖ Jobs in the steel industry ❖ Number of employees in a firm
Macroeconomics	❖ National Production/Output ❖ Total Industrial Output ❖ Gross Domestic Product ❖ Growth of Output	❖ Aggregate Price Level ❖ Consumer prices ❖ Producer Prices ❖ Rate of Inflation	❖ National Income ❖ Total wages and salaries ❖ Total corporate profits	❖ Employment and Unemployment in the Economy ❖ Total number of jobs ❖ Unemployment rate

Introduction to Engineering Economics

Economic Institutions

- To apply economic theory to reality, you've got to have a sense of economic institutions.
- Economic institutions are laws, common practices, and organizations in a society that affect the economy
- Economic institutions differ significantly among nations
- They sometimes seem to operate differently than economic theory predicts

Economic Policy Options

- Economic policies are actions (or inactions) taken by the government to influence economic actions
- Objective policy analysis keeps value judgments separate from the analysis (An "objective policy" refers to a policy that is based on facts, data, and measurable outcomes, rather than personal opinions or beliefs)
- Subjective policy analysis reflects the analyst's views of how things should be ("subjective policy" refers to a policy process or evaluation that considers individual perceptions, values, and interpretations, rather than solely relying on objective data or facts)

Introduction to Engineering Economics

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Introduction to Engineering Economics

Economic Terminology

- ✓ Goal of this class is to describe how economics works in the real world.
- ✓ You will be introduced to many terms that occur in business and in discussions of the economy.

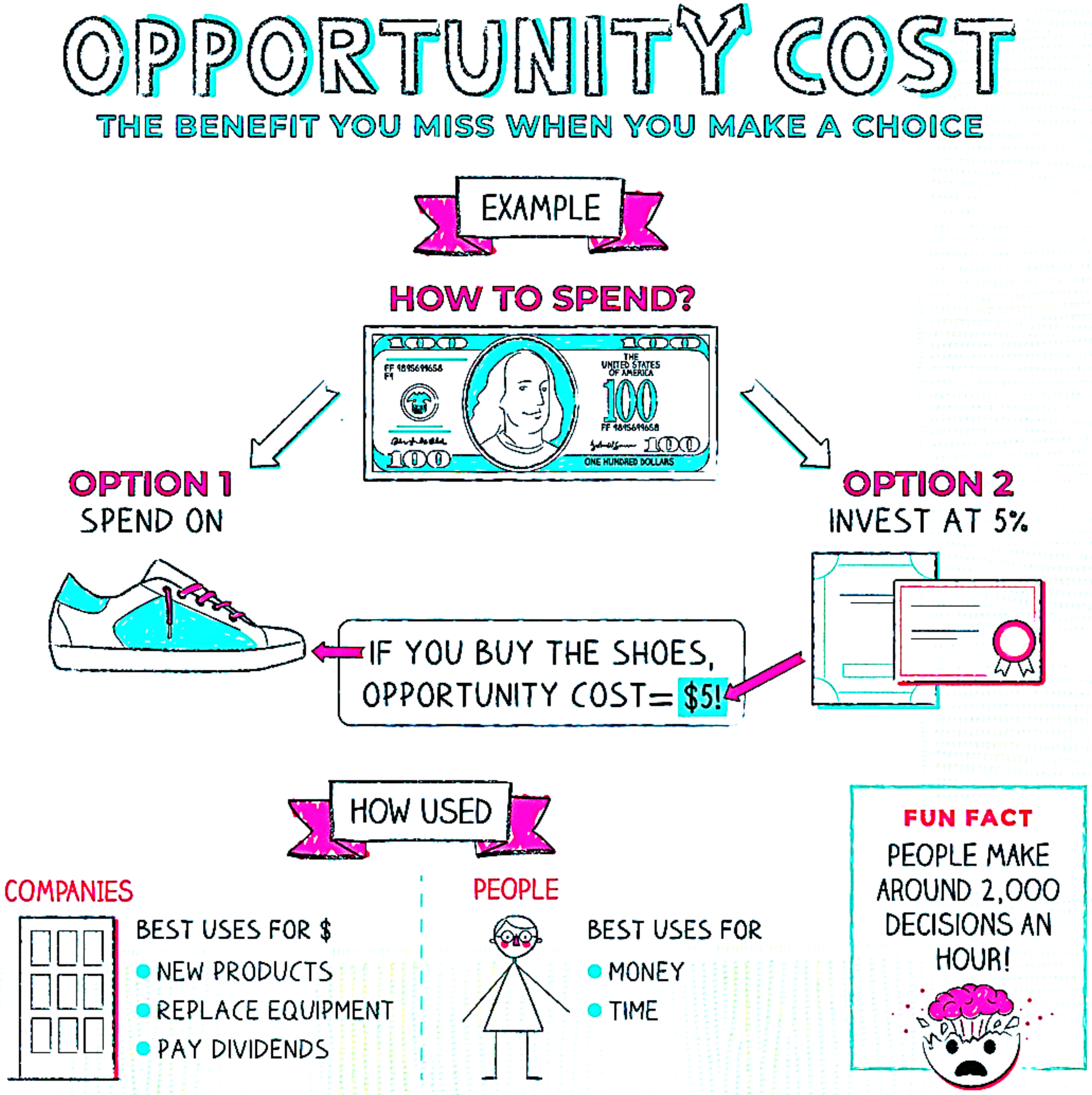
- ❖ Opportunity cost
- ❖ Marginal benefit and marginal cost
- ❖ Market and economic forces
- ❖ Demand Supply
- ❖ Production Function
- ❖ Optimization, etc.

Opportunity cost

- ✓ Opportunity cost is the benefit forgone of the next-best alternative to the activity you have chosen
- ✓ Opportunity cost should always be less than the benefit of what you have chosen
- ✓ Opportunity cost is the basis of cost/benefit economic reasoning

Marginal Costs and Marginal Benefits

- ✓ Using economic reasoning, decisions are often made by comparing marginal costs and marginal benefits
- ✓ Marginal cost is the additional cost over and above costs already incurred
- ✓ Marginal benefit is the additional benefit above and beyond what has already accrued



Napkin Finance

Opportunity Cost Formula

$$\textbf{Opportunity Cost} = \text{Total Revenue} - \text{Economic Profit}$$

$$\textbf{Opportunity Cost} = \frac{\text{What One Sacrifice}}{\text{What One Gain}}$$

Introduction to Engineering Economics

Marginal benefit is the benefit from each additional increment or unit.

Marginal Cost is the cost from each additional increment or unit.

Hours Studied	Exam Grade
0	30%
1	65%
2	85%
3	95%
4	100%
5	95%

If an economics student wanted to maximize their total benefit, they would study 4 hours. The total benefit of studying 4 hours is a grade of 100% on the exam. No other amount of studying would allow this student to get such a high grade?

Another misconception many non-economists have is that people should work to maximize their total benefits.

Introduction to Engineering Economics

Hours Studied	Exam Grade
0	30%
1	65%
2	85%
3	95%
4	100%
5	95%

Hours Studied	Total Benefit	Marginal Benefit
0	$30	N/A
1	$65	$35
2	$85	$20
3	$95	$10
4	$100	$5
5	$95	-$5

Marginal Benefit simply means change in the total Benefit for every each hour .

Generally Decreases for Every Each Hour

Another misconception many non-economists have is that people should work to maximize their total benefits.

Introduction to Engineering Economics

Hours Studied	Exam Grade
0	30%
1	65%
2	85%
3	95%
4	100%
5	95%

Hours Studied	Total Cost	Marginal Cost
0	$0	N/A
1	$6	$6
2	$13	$7
3	$22	$9
4	$34	$12
5	$49	$15

Marginal Cost simply means change in the total Cost for every each hour .

Generally Increase for Every Each Hour

Another misconception many non-economists have is that people should work to maximize their total benefits.

Introduction to Engineering Economics

$$MC = \frac{\Delta C}{\Delta Q}$$

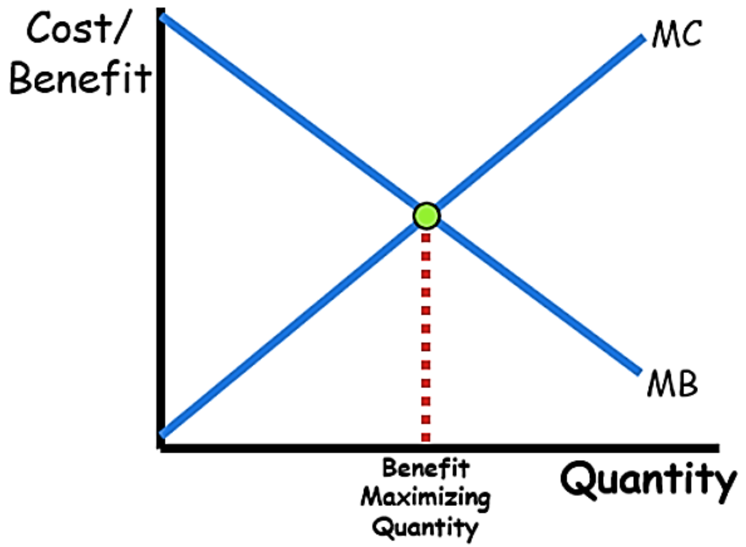
where,

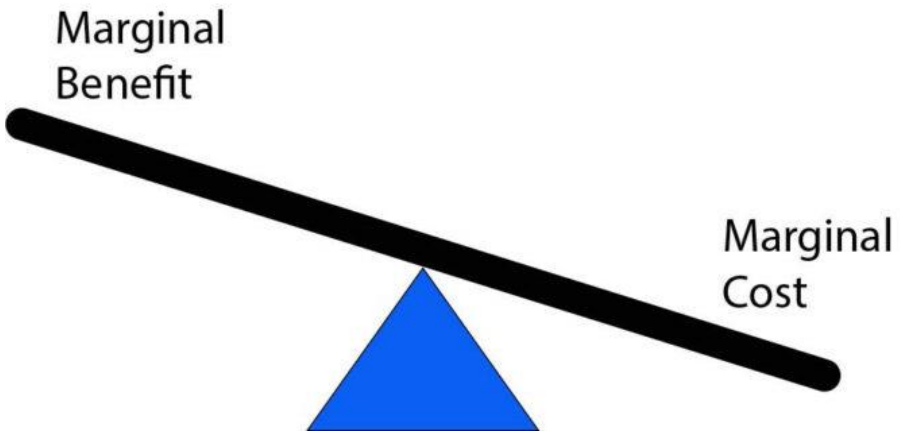
MC = Marginal Cost

ΔC = change in cost of production

ΔQ = change in quantity

Number of Uniforms (Output)	Total Cost (in $)	Marginal Cost (ΔC/ΔQ)
5	68	68/5 = $13.6
6	76	(76-68)/(6-5) = 8/1 = $8
8	88	(88-76)/(8-6) = 12/2 = $6
9	96	(96-88)/(9-8) = 8/1 = $8
10	110	(110-96)/(10-9) = 14/1 = $14





Introduction to Engineering Economics

The economic decision rule

If the marginal benefits of doing something exceed the marginal costs, do it.

MB > MC → Do it!

If the marginal costs of doing something exceed the marginal benefits, don't do it.

MC > MB → Don't do it!

Examples of opportunity cost:

1. **Individual decisions:**

The opportunity cost of college includes: Items you could have purchased with the money spent for tuition and books Loss of the income from a full-time job.

2. **Government decisions:**

The opportunity cost of money spent on the war on terrorism is less spending on health care or education.